

Interactive 3D Medical Image Segmentation Software

User Manual

Overview

This interactive 3D medical image segmentation software is designed to integrate seamlessly into the medical imaging workflow and can play a pivotal role in enhancing diagnosis, monitoring, and also treatment plans of patients. Its primary purpose is to facilitate detailed and accurate segmentation of medical images (including MR and CT images) using artificial intelligence techniques that do not require extensive expert training and guidance.

The software has been thoughtfully designed for use by clinicians, such as oncologists and radiologists, in addition to imaging technicians and reduces the overhead of medical image segmentation - giving you more time for crucial aspects such as treatment planning.

Major functionalities include:

- Compatibility with standard medical imaging formats; DICOM and NIfTI images.
- Perform segmentation across axial, coronal, and sagittal views to capture comprehensive anatomical information.
- Utilise segmentation at various resolutions to adapt to different clinical requirements and levels of detail.
- Support segmentation of multiple objects simultaneously and enable cross-view integration for thorough spatial context.
- Leverage immersive 3D visualisation to enhance understanding of complex anatomical structures.
- Easily export segmentation results for further analysis, reporting, or integration into downstream clinical systems.

Furthermore, the software offers flexibility in processing, which can be conducted locally, or remotely, if limited hardware is available.

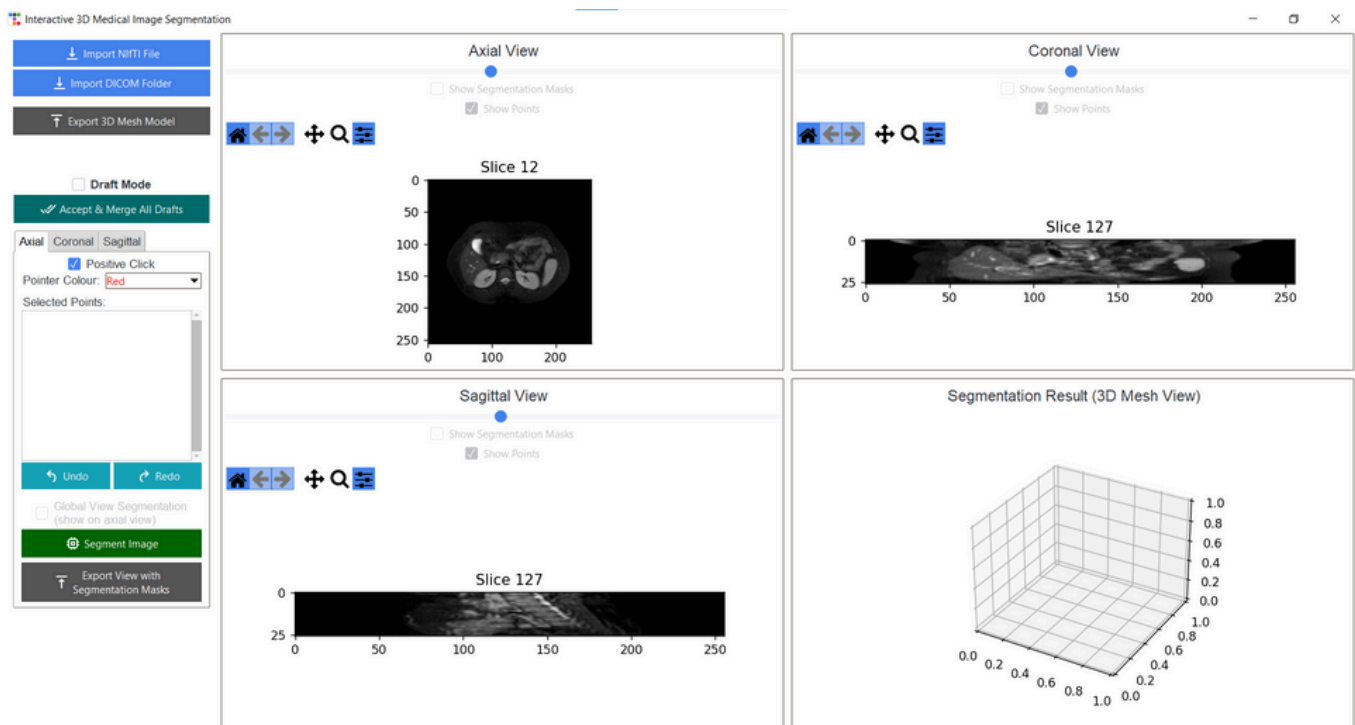
Hardware Requirements

A machine (running Windows 10 or Windows 11 operating system) is required to run the graphical user interface.

To run the segmentation model locally; a CUDA-supported GPU (version 12.2 or later) is required.

To run the segmentation model remotely; a stable internet connection and Modal serverless instance is required. To use Modal, an account is necessary (sign-up can be completed via <https://modal.com/signup>).

User Interface and Instructions



After starting the program and importing a valid DICOM/NIfTI image (via buttons in the top left corner of the screen), the program will appear like above. Under the import buttons there are tabs (axial, coronal, sagittal) to control how the corresponding plot (on the right of the screen) is manipulated.

Importantly, a “Draft Mode” feature exists to experiment with segmentation results before committing the results to a more-refined, final version. Draft mode is particularly useful when you want to use multiple slices in the same view as the start point for independent segmentations (for example, when performing multiorgan segmentation with occlusions that prevent segmenting all organs in the same slice). Draft mode can be entered by checking the checkbox with the same name in the sidebar. Whether in or out of draft mode, all of the features below are supported:

In each tab;

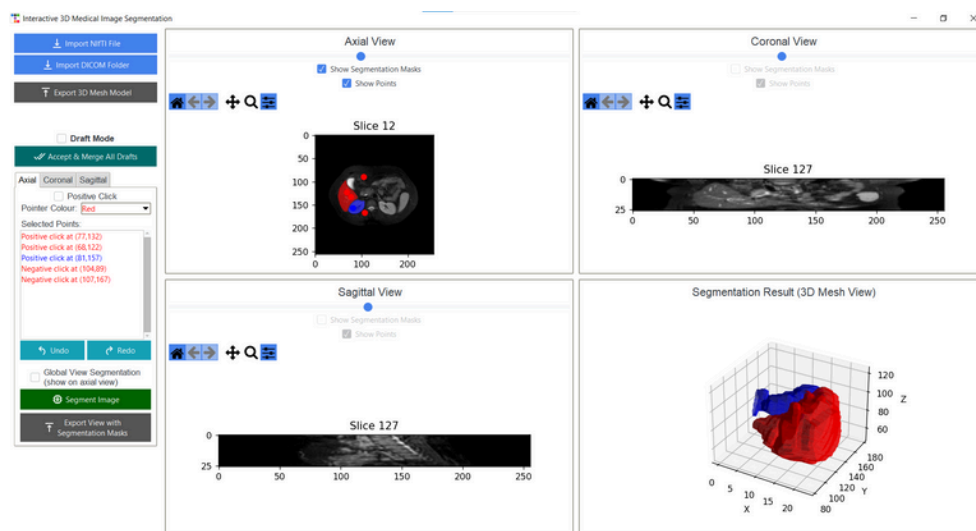
There is an option to choose a pointer colour (each pointer colour corresponds to a different object/region of interest that is to be segmented).

As well as the option to make positive and negative clicks - positive clicks encourage the the segmentation model to expand the mask and negative clicks constrain masks (for this reason, negative clicks are useful correction tools).

To make a selection, simply click on the desired area of the corresponding plot on the right side of the screen. The type of click (positive/negative) and coordinates of the selected point will show in the listbox below the pointer colour option.

You may make any number of positive and negative clicks - if you make any mistakes, simply click “Undo”.

After making the desired selections, you may segment the image (by clicking the button at the bottom of the tab).



After completing segmentation, you will see a screen similar to the screenshot above. Segmentation masks have been overlayed on the respective view and the 3D mesh view (discussed below) has been generated. You can track the segmentation mask throughout the view by moving the slider at the top of the respective view in the grid on the right of the screen. You may also use the zoom and pan tools (located under the slider) to closely inspect the segmentation results. Also located under the slider are the options to hide/show both segmentation masks and points for cleaner evaluation of the quality of the segmentation mask.

If you were previously in draft mode, you can now accept and merge your results from the draft segmentation and keep them. NOTE: if you have performed segmentation using the same colour (ie on the same object), accepting and merging results from draft segmentation will overwrite your existing segmentations with the draft segmentations.

The 3D mesh view can be rotated for better visualisation. It tracks and display the 3D mesh model of all segmentation masks across all frames (for this reason, the 3D mesh view in draft mode is distinct from the 3D mesh view used when the user is out of draft mode).

You may repeat segmentation on the current view; or switch views by changing tabs and then changing the pointer colour etc.

Global view segmentation (accessible via a checkbox with the same name above the “Segment Image” button) allows users to select points in coronal and sagittal views and see the resulting segmentation mask in axial view. It is important that the axial slice index corresponds to at least one y-coordinate of a selection in coronal or sagittal view. Once finished with global view segmentation, please remember to uncheck the box to prevent unwanted masks appearing in axial view.

At any point, segmentation results corresponding to the view can be exported as binary, grayscale or RGB NIfTI files, by clicking the last button at the bottom of the tab. Exporting the generated 3D mesh as an STL file is also supported at any point, by clicking Export 3D Mesh View button in the top left corner of the application.

FAQ

Q1: I am choosing a region of interest but the segmentation mask is “too general”. How can I solve this?

A1: There are two ways to solve this issue;

- 1) Add more positive clicks within your region of interest.
- 2) Add negative clicks outside of your region of interest (in undesired areas of the segmentation output).

Q2: Is remote or local processing faster?

A2: While both local and remote processing is quick and convenient, local processing is typically slightly faster; this is due to network bandwidth limitations (when communicating with the remote segmentation model) and the overhead of configuring the environment for each request. We will continue to make optimisations to reduce remote processing times.

Glossary

- **DICOM:** The standard format for storing, transmitting, and handling medical images and related information.
- **NIFTI:** A file format widely used in neuroimaging that stores volumetric imaging data, facilitating analysis and compatibility across platforms.
- **STL:** A widely used file format for representing 3D models.
- **MRI:** A non-invasive imaging technique that uses magnetic fields and radio waves to produce detailed images of soft tissues.
- **CT:** An imaging method that uses X-rays to create detailed cross-sectional (slice) images of the body.
- **Medical Image Segmentation:** The process of partitioning an image into meaningful regions or segments to isolate anatomical structures for further analysis.
- **Multiresolution Segmentation:** A method that analyzes images at various resolution levels to balance detail and contextual information.
- **Cross-view Segmentation:** A strategy that integrates segmentation data from multiple anatomical planes—axial, coronal, and sagittal—to create cohesive 3D segmentations.
- **Artifact:** Any distortion or error in imaging data that may obscure or alter the appearance of true anatomical structures.
- **Positive Click:** An interactive input where users mark areas within the ROI to affirm inclusion in the segmentation process.
- **Negative Click:** An interactive input where users indicate areas outside the desired ROI to help exclude them from the segmentation.

Changelog

Version 1.0 (April 2025): Initial version

Support Information

This software has been developed by Sener Bradley Topaloglu (psyst6@nottingham.ac.uk).